

Operators:

Common

+	(1,3)	Plus (also for string)	-	(1,3)	Minus
++		Increment pre and post (++V, V++)	--		Decrement pre and post (--V, V--)
!	(4)	Not	~	(2,3)	Complement
*	(1,3)	Multiplication	/	(1)	Division
%		Remainder	**	(1,3)	Power
&&		Logical AND	&		AND
		Logical OR			OR
<<		Shift left	>>		Shift right
>>>		Unsign shift right	#		XOR
<		Less	>		Greater
!=	(1,3)	Not equal	<>	(1,3)	Not equal
+=		Increase	-=		Decrease
*=		Mul and assign	/=		Div and assign
%=		Rem and assign	**=		Pow and assign
<<=		Shift and assign	>>=		Shift and assign
#=		XOR and assign	&=		AND and assign
^=		XOR and assign	>>=		Shift and assign

'Pascal' - style

:=	(1,3)	Assign	=	(1,3)	Compare
^	(1,3)	Power	//	(1,3)	Parallel resistors

'C' - style

=	(1)	Assign	==	(1,3)	Compare
^		XOR	//	(1,3)	Parallel resistors

(1) Also for complex operands. (2) Complex conjugate: $\sim(1+2i)=1-2i$

(3) Also for matrix operands. (4) Matrix inversion.

First using of '==' switches engine to 'C'-style syntax.

First using of ':' switches engine to 'Pas'-style syntax.

Default is 'Pas'-style syntax.

Functions:

<code>abs(x z M)</code>	(1,3)	Absolute value Complex Matrix
<code>sin(x z)</code>	(1)	Sin of radian Complex
<code>sing(x)</code>		Sin of degrees
<code>cos(x z)</code>	(1)	Cos of radian Complex
<code>cosg(x)</code>		Cos of degrees
<code>tg(x z), tan(x z)</code>	(1)	Tangent of radian Complex
<code>tgg(x)</code>		Tangent of degrees
<code>ctg(x z), cot(x z)</code>	(1)	Cotangent of radian Complex
<code>ctgg(x)</code>		Cotangent of degrees
<code>arcsin(x z), asin(x z)</code>	(1)	ArcSin in radian Complex
<code>arccos(x z), acos(x z)</code>	(1)	ArcCos in radian Complex
<code>arctg(x z), atan(x z)</code>	(1)	ArcTg in radian Complex
<code>arcctg(x z), acot(x z)</code>	(1)	ArcCtg in radian Complex
<code>atan2(y z,x z)</code>	(1)	Arc tangent of y/x.
<code>sinh(x z), sh(x z)</code>	(1)	Hyperbolic sine
<code>cosh(x z), ch(x z)</code>	(1)	Hyperbolic cosine
<code>tanh(x z), th(x z)</code>	(1)	Hyperbolic tangent
<code>ctanh(x z), cth(x z)</code>	(1)	Hyperbolic cotangent
<code>asinh(x z), arsh(x z)</code>	(1)	Hyperbolic arc sine
<code>acosh(x z), arch(x z)</code>	(1)	Hyperbolic arc cosine
<code>atanh(x z), arth(x z)</code>	(1)	Hyperbolic arc tangent
<code>acoth(x z), arcth(x z)</code>	(1)	Hyperbolic arc cotangent
<code>exp(x z)</code>	(1)	Exponent, e^x
<code>exp10(x z)</code>	(1)	10^x
<code>pow(x z,y z)</code>	(1)	x^y
<code>ln(x z), log(x z)</code>	(1)	Natural logarithm
<code>lg(x z), log10(x z)</code>	(1)	Logarithm base 10, $\text{Log}_{10}(x)$
<code>log2(x z)</code>	(1)	Logarithm base 2, $\text{Log}_2(x)$
<code>logn(x z,y z)</code>	(1)	$\text{Log } x(y)$
<code>sqrt(x z) root2(x z)</code>	(1)	Square root
<code>root3(x z), cbrt(x z)</code>	(1)	Cube root, $x^{(1/3)}$
<code>rootn(x z,n z)</code>	(1)	nth root, $x^{(1/n)}$
<code>hypot(x z,y z)</code>	(1)	Hypotenuse
<code>round(x z)</code>	(1)	Round value
<code>int(x z)</code>	(1)	Convert to int
<code>float(x z)</code>	(1)	Convert to float (do nothing)
<code>ceil(x z)</code>	(1)	Rounds the number x up
<code>floor(x z)</code>	(1)	Rounds the number x down
<code>frac(x z)</code>	(1)	Returns the fractional part
<code>not(n)</code>		Inversion all bits, $\sim n$
<code>gcd(n,m)</code>		Greatest common divisor of n and m
<code>invmod(n,m)</code>		Inverse of n modulo m
<code>mod, fmod(x z,y z)</code>	(1)	Reminder
<code>rnd(x z)</code>	(1)	Random value 0..x

erf(x)		Error function
erfc(x)		Complementary error function
db(x z)	(1)	Decibel, $10*\lg(x)$, $10*\lg(\text{abs}(z))$
np(x z)	(1)	Neper, $20*\lg(x)$, $20*\lg(\text{abs}(z))$
adb(x z)	(1)	$10^{(x/10)}$, $10^{(z/10)}$
anp(x z)	(1)	$10^{(x/20)}$, $10^{(z/20)}$
Vout(Vref,Rh,Rl)		$V_{\text{out}}=V_{\text{ref}}*(R_h+R_l)/R_l$
frh(x)		Fahrenheit to Celsius.
ee(n,x)		E series of preferred numbers $ee(24, 5k)=5.1k$
cs(x)		Cross section of diameter
acs(x)		Diameter of cross section
awg(n)		American wire gauge to mm
sawg(n)		American wire gauge to mm ²
aawg(x)		mm to American wire gauge
swg(n)		British standard wire gauge to mm
sswg(n)		British standard wire gauge to mm ²
aswg(x)		mm to British standard wire gauge
polar(r,θ)		Return complex value $r*\exp(i\theta)$
cmplx(x,y), cplx(x,y) cpx(x,y)		Return complex value $x+iy$
re(z)	(1)	Return real part of complex
im(z)	(1)	Return imaginary part of complex
conj(z)	(1)	Complex conjugate
pol(z)	(1)	$\text{atan}(\text{im}(z)/\text{re}(z))$
min(x,y)		Minimum of x, y
max(x,y)		Maximum of x, y
fact(n x z), n x z!	(1)	Factorial (n!)
prime(n)		Smallest prime number
winf(x)		Returns a string describing the portion of the spectrum of a given wavelength.
wrgb(x)		Returns the RGB color of the given wavelength in 0xRRGGBB format.
trgb(x)		Returns the RGB color of the given absolute temperature in 0xRRGGBB format.
color(n)		Shows a color (colored square on top of the application in 0xRRGGBB format)
const("name",x z M)	(1,3)	Add (declare) constant
var("name",x z M)	(1,3)	Add (declare) variable
integr(expr, from, to, x)		Numeric Gauss-Kronrod integration
diff(expr, point, x)		Numeric differentiation
sum(expr, from, to, x)		$\sum \text{expr}(x)$
for(expr, from, to, x)	(1,3)	Repeat expr calculations for x and return last expr value.
if(cond, true, false)	(1,3)	Return true if $\text{cond} \neq 0$ else return false.
tr(A)	(3)	trace = sum of diagonal

det(A)	(3)	Determinant of matrix
norm(A)	(3)	Frobenius norm
dot(A, B)	(3)	Dot product of two vectors (1×N or N×1), any length
cross(A, B)	(3)	Cross product of two 3-element vectors (1×3 or 3×1)
rows(A)	(3)	Return matrix rows
cols(A)	(3)	Return matrix columns
size(A)	(3)	Return matrix rows*columns
zeros(r,c M)	(3)	Return zero matrix with r rows and c columns
diag(r,c M), eye(r,c M)	(3)	Return zero matrix with r rows and c columns with 1 diagonal
A[r, c] , A[i]	(3)	Matrix element access (both read and write) to the matrix element
bind(x)		Reinterpret a floating-point value as an integer 64
floatd(i)		Reinterpret an integer as a floating-point 64
binf(x)		Reinterpret a floating-point value as an integer 32
floatf(i)		Reinterpret an integer as a floating-point 32
printf(fmt, ...),fprn, prn	(1)	C-style printf. Result in the string value.
prnf(filename, fmt, ...)		Works the same as prn, but print to the filename.
datetime("2017.11.23 22:00:20")		Return Unix time
now(n)		Current Unix time (+/- n hours)

Legend: x, y - real number; z - complex number; n, m, i - integer number; A, B, M - matrix; c, r - column, row of matrix

(1) Also for complex operands.

(3) Also for matrix operands.

Special:

help(0)	Call help
opacity(100)	Set opacity level in percent
binwide(32)	Set binary width
menu(0)	Hide (0) or show (1) menu.
vars(0)	Show variables list
font(x)	Set calculator font size.
prec(n)	Set decimal digits in output.

User defined functions:

```
{frq(L, C)1/(2 pi sqrt(L C))} (1,3)    frq(130u, 2n2)
```

Newton-Raphson solution of the equation:

```

solve(x(2x+2)-2, x:=0) (1) 0.6180339887500326
solve(z^2+z+1, z:=0+1i) (1) |1|(120`0'0") -500m+866mi
solve(x(2x+2)-2, x:=-1) (1) -1.61803398875005
calc(x(2x+2)-2, x:=-1) (1) -2
calc(z^2+z+1, z:=0+1i) (1) |1|(90`0'0") 0+1i
solve(expr, var:=est) expr is the left side of the equation,
expr(var)=0, est is the first approximation of the root.

```

Data formats:

Hex	0x1000, \$1000	4096
Octal	01000, 0o1000	512
Binary	0b1000	8
Degrees	10`20'40"	0.180544 rad.
Date time	1:c1:y1:d1:h1:m1:s	189377247661s
Char	'c'	99
WChar	L'c', 'c'W	'c'W
String	'string'+ " value"	"String value"
Other	\377 = 255, \e = 27, \n = 10, \xff = 255	
Normalized	1.8e6	1800000
Scientific	1.8M	1800000
Engineering	1k33	1330
Computing	1KB, 1KiB	1024
Fractional	3+9/64	Pi (best rational approximation)
Gauge	#28, awg(28)	0.3211265489669004
Inch	2"5	0.0635=2+1/2 "
Complex	3+4i, 3+j4, j4+3	3+4i
Temperature	frh(75), 75F	297.039 K 23.8889 `C 75 `F
Matrix	A := [(1,2,3);(4,5,6);(7,8,9)];C := [(1,2);(3,4);(5,6)]	

Examples:

L:=130u;c:=2.2n;f:=1/(2*pi*sqrt(1*c));;Resonant	297.6k
72-20%	57.6
72-20%+5%	60.48
500*5%	12500
1/2%	50
57.6%%72	-20
abs(3+4j);;Comment	5
z:=3+4i	3+4i
integr(1/sqrt(x), 0.001, 1, x)	1.93675444679663
time:=2*pi;sqrt((1/time)integral((10sin(x))^2, 0, time, x))	7.071067811865472
diff(tg(x), 0, x)	1
sum(1/x!,20,0,x)	2.718281828459045
A := [(1,2,3);(4,5,6);(7,8,9)];tr(A)	15
A := [(1,2,3);(4,5,6);(7,8,9)];norm(A)	16.881943016134134
A := [(1,2,3);(4,5,6);(7,8,9)];cols(A)	3

Implicit Multiplication:

3PI	3*PI	9.424777960
3pi	0+3pi	0+3e-12i
2 pi	2*pi	6.283185307
2sin(2PI/3)	2*sin(2*PI/3)	1.732050807
3(1+2)	3*(1+2)	9
(1+2)(1+2)(1+2)	(1+2)*(1+2)*(1+2)	27

Constants:

version	2.02 or above
pi	3.141592653589
e	2.718281828459
phi	1.618033988749 (Golden ratio)
timezone	System timezone offset (hours)
daylight	Daylight saving time flag
tz	Offset from UTC in hours

Range postfix:

Binary (power of 2)

1QB	quetta	1267650600228229401496703205376	2 ¹⁰⁰
1RB	ronna	1237940039285380274899124224	2 ⁹⁰
1YB	yotta	1208925819614629174706176	2 ⁸⁰
1ZB	zetta	1180591620717411303424	2 ⁷⁰
1EB	exa	1152921504606846976	2 ⁶⁰
1PB	peta	1125899906842624	2 ⁵⁰
1TB	tera	1099511627776	2 ⁴⁰
1GB	giga	1073741824	2 ³⁰
1MB	mega	1048576	2 ²⁰
1KB	kilo	1024	2 ¹⁰

Decimal (power of 10)

			EU	US			
1Q	quetta	10 ³⁰	Quintillion	Nonillion	1q	quecto	10 ⁻³⁰
1R	ronna	10 ²⁷	Quadrilliard	Octillion	1r	ronto	10 ⁻²⁷
1Y	yotta	10 ²⁴	Quadrillion	Septillion	1y	yocto	10 ⁻²⁴
1Z	zetta	10 ²¹	Trilliard	Sextillion	1z	zepto	10 ⁻²¹
1E	exa	10 ¹⁸	Trillion	Quintillion	1a	atto	10 ⁻¹⁸
1P	peta	10 ¹⁵	Billiard	Quadrillion	1f	femto	10 ⁻¹⁵
1T	tera	10 ¹²	Billion	Trillion	1p	pico	10 ⁻¹²
1G	giga	10 ⁹	Milliard	Billion	1n	nano	10 ⁻⁹
1M	mega	10 ⁶	Million	Million	1u	micro	10 ⁻⁶
1K	kilo	10 ³	Thousand	Thousand	1m	milli	10 ⁻³
1k	kilo	10 ³	Thousand	Thousand			
1h	hecto	10 ²	Hundred	Hundred	1c	centi	10 ⁻²
1D	deka	10 ¹	Ten	Ten	1d	deci	10 ⁻¹
1da	deka	10 ¹	Ten	Ten			
1		10 ⁰	One	One	1		10 ⁰

Predefined constants:

Name	Value	Description
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Mathematical:

pi	3.141592653589793	π
tau	6.283185307179586	2π
turn	6.283185307179586	2π
gon	0.01570796326794897	$\pi/200$
deg	0.0174532925199433	$\pi/180$
e	2.718281828459045	Euler's number
i, j	0+1i	$\sqrt{-1}$
phi	1.618033988749895	Golden ratio

Imperial and other:

Distance:

inch	0.0254	Inch (m)
mil	2.54e-05	1/1000Inch (m)
ft	0.3048	Foot (m)
foot	0.3048	Foot (m)
yd	0.9144	Yard (m)
yard	0.9144	Yard (m)
mi	1609.344	Mile (m)
mile	1609.344	Mile (m)
nmi	1852	Nautical mile (m)
ptt	0.0003527777777777778	Printing point DTP 0.3528mm

Mass:

kg	1	Kilogram (kg)
lb	0.45359237	Pound (kg)
oz	0.028349523125	Ounce (kg)
st	6.35029318	Stone (kg)
gr	0.001	Gram (kg)

Volume:

gal	0.003785411784	US Gallon (m ³)
qt	0.000946352946	US Quart (m ³)
pt	0.000473176473	US Pint (m ³)
cup	0.0002365882365	US Cup (m ³)
floz	2.95735295625e-05	US Fluid Ounce (m ³)
tbsp	1.478676478125e-05	US Tablespoon (m ³)
tsp	4.92892159375e-06	US Teaspoon (m ³)
lt	0.001	liters (m ³)
ml	1e-06	milliliters (m ³)
cc	1e-06	milliliters (m ³)

Energy:

cal	4.184	Calorie (J)
kcal	4184	Kilocalorie (J)
btu	1055.05585262	British thermal unit (J)
wh	3600	Watt hour (J)
kwh	3600000	Kilowatt hour (J)
mwh	3600000000	Megawatt hour (J)
gtnt	4184	Gram of TNT (Joules)
ttnt	4184000000	Tonne of TNT (Joules)
ktnt	4184000000000	Kiloton of TNT (Joules)
mtnt	4184000000000000	Megaton of TNT (Joules)
ev	1.602176634e-19	Electronvolt (J)
kev	1.602176634e-16	Kiloelectronvolt (J)
mev	1.602176634e-13	Megaelectronvolt (J)
gev	1.602176634e-10	Gigaelectronvolt (J)

Power:

hps	745.6998715822702	Horsepower (W)
bhp	745.6998715822702	Brake horsepower (W)

Pressure:

atm	101325	Standard atmosphere (Pa)
bar	100000	Bar (Pa)
psi	6894.757293168361	Pound-force per square inch (Pa)

Velocity:

kmh	0.277777778	Kilometers per hour to meters per second
mph	0.44704	Miles per hour to meters per second
knot	0.514444444	Nautical miles per hour to meters per second

Time:

hour	3600	Hour in seconds
hr	3600	Hour in seconds
mnt	60	Minute in seconds

Magnetic:

tl	1	Magnetic flux density (or magnetic induction)
Tesla - Base SI		
wb	1	Magnetic flux Weber - Base SI
gs	0.0001	Magnetic flux density (or magnetic induction)
Gauss to Tesla		
mw	1e-08	Magnetic flux Maxwell to Weber
oe	79.57747154594767	Magnetic field strength (H) Oersted to A/m
gb	0.7957747154594768	Magnetomotive force (MMF) Gilbert to Ampere-
turn		

Radiation:

gy	1	Gray (J/kg) - Base SI
rad	0.01	Rad (absorbed dose)
sv	1	Sievert (Equivalent dose) - Base SI
rem	0.01	Roentgen Equivalent Man (0.01 Sv)
rn	0.00877	Roentgen (approx in air)
mrn	0.00000877	Milli-roentgen
urn	8.77e-09	Micro-roentgen
ngnt	0.031572	Not great, not terrible (3.6 R)

Physical Constants (CODATA 2018):

Fundamental:

c0	299792458	Speed of light (m/s) The speed of
electromagnetic radiation in vacuum		
hp	6.62607015e-34	Planck constant (J·s) Relating energy to
frequency		
hb	1.054571817e-34	Reduced Planck constant $\hbar$ (J·s) Planck
constant divided by 2π		
gn	6.6743e-11	Gravitational (Newton's) constant ($\text{m}^3/(\text{kg}\cdot\text{s}^2)$)
na	6.02214076e+23	Avogadro constant (mol^{-1}) Number of
constituent particles per mole		
kb	1.380649e-23	Boltzmann constant (J/K) Relates average
kinetic energy of particles to temperature		
rg	8.314462618	Universal gas constant (J/(mol·K)) Product of
Boltzmann constant and Avogadro constant		
sf	5.670374419e-08	Stefan-Boltzmann constant ($\text{W}/(\text{m}^2\cdot\text{K}^4)$)
rs	8.314462618	Ideal gas constant $k_b\cdot N_a$ (J/(mol·K))
nae	2.1798723611035e-18	Rydberg energy (J)
mu	1.6605390666e-27	Atomic mass constant (kg)
stdt	273.15	Standard temperature 0°C (K)

Electromagnetic:

e0	8.8541878128e-12	Electric constant (vacuum permittivity) (F/m)
Permittivity of free space		
u0	1.25663706212e-06	Magnetic constant (vacuum permeability) (H/m)
Permittivity of free space		
z0	376.730313668	Impedance of vacuum (Ω) Wave impedance of
electromagnetic waves in vacuum		

Particle:

qe	1.602176634e-19	Elementary charge (C) Electric charge of a
proton (magnitude)		
me	9.1093837015e-31	Electron mass (kg) Rest mass of an electron
mp	1.67262192369e-27	Proton mass (kg) Rest mass of a proton
mn	1.67492749804e-27	Neutron mass (kg) Rest mass of a neutron
rel	2.8179403262e-15	Classical electron radius (m) Characteristic
length scale for electron scattering		
a0	5.29177210903e-11	Bohr radius (m) Most probable distance between
nucleus and electron in hydrogen ground state		

Astronomical:

au to Sun	149597870700	Astronomical unit (m) Mean distance from Earth
ly Julian year	9460730472580800	Light year (m) Distance light travels in one
pc astronomical distance unit	3.0856775814914e+16	Parsec (m) Parallax of one arcsecond,
g0	9.80665	Standard gravity (m/s ²)

Additional:

ry for atomic spectra	10973731.56816	Rydberg constant (m ⁻¹) Spectroscopic constant
sb Constant relating black body radiation to temperature	5.670374419e-08	Stefan-Boltzmann constant (W/(m ² ·K ⁴))

Rainbow Colors:

Wavelengths (in nanometers, nm)		
fir	316e-6	Far infrared (FIR, 100-1000 um)
lwir	11.5e-6	Long-wave IR (LWIR, 8-15 um, thermal)
mwir	5.5e-6	Mid-wave IR (MWIR, 3-8 um, night vision)
swir	2.2e-6	Short-wave IR (SWIR, 1.4-3 um, fiber)
nir	1.09e-6	Near IR (NIR, 0.78-1.4 um, remote)
red	685e-9	(~620-750 nm)
orange	605e-9	(~590-620 nm)
yellow	580e-9	(~570-590 nm)
green	532e-9	(~495-570 nm)
blue	472e-9	(~450-495 nm)
indigo	435e-9	(Often grouped with blue/violet, ~420-450 nm)
violet	400e-9	(~380-450 nm)
uva	348e-9	UVA ultraviolet (315-380 nm, tanning)
uvb	298e-9	UVB ultraviolet (280-315 nm)
uvc	190e-9	UVC ultraviolet (100-280 nm, germicidal)

Integer Limits:

max32, maxint	Maximum signed 32-bit (2147483647, 0x7FFFFFFF)
maxu32, maxuint	Maximum unsigned 32-bit (4294967295, 0xFFFFFFFF)
max64, maxlong	Maximum signed 64-bit (9223372036854775807, 0x7FFFFFFFFFFFFFFF)
maxu64, maxulong	Maximum unsigned 64-bit (18446744073709551615, 0xFFFFFFFFFFFFFFFF)

Floating point parameters:

fp_dig	Significant decimal digits
fp_mant_dig	Bits in the mantissa
fp_max_exp	Maximum exponent (base 2)
fp_max_10_exp	Maximum exponent (base 10)
fp_sz	Size of floating point number in bits
int_sz	Size of integer number in bits

System:

version	Calculator version
timezone	System timezone offset in hours
daylight	Daylight saving time flag (0 or 1)
tz	Current timezone with DST applied

Note: All constants are case-insensitive without Case sensitive option.